



IPHONE SE INVESTIGATIVE REPORT

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Table of Contents

CHAPTER ONE: INTRODUCTION	2
1.1 Background of the Study	2
1.2 Problem Statement	2
1.3 Objectives of the Study	2
1.4 Scope of the Project	3
1.5 Significance of the Study	3
CHAPTER TWO: METHODOLOGY	4
2.1 Overview of the Forensic Process	4
2.2 Tool Selection and Justification	4
2.3 Examination Process Using Magnet AXIOM	5
2.4 Supplementary Examination with Other Tools	5
2.4.1 MOBILedit Forensic Express	5
2.4.2 Autopsy	5
2.5 Analytical Procedure	6
CHAPTER THREE: FINDINGS AND ANALYSIS	7
3.1 Findings and Question-Based Analysis	7
3.2 Overall Analysis and Discussion	17
CHAPTER FOUR: CHALLENGES, CONCLUSION AND RECOMMENDATIONS	17
4.1 Challenges Encountered	17
4.1.1 Limited Image Format	17
4.1.2 Incompatibility with Some Forensic Tools	17
4.1.3 Partial Artifact Availability	18
4.1.4 Absence of Encrypted Data	18
4.1.5 Time and Resource Constraints	18
4.2 Conclusion	18
4.3 Recommendations	19

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The increasing reliance on smartphones for communication, transactions, and storage of personal data has made them prime sources of digital evidence in modern forensic investigations. Mobile devices, particularly those running Apple's iOS, store extensive information such as call logs, messages, media, application data, and system configuration files that can be invaluable in both criminal and corporate investigations. However, due to Apple's emphasis on privacy and security, the extraction and analysis of such data pose unique challenges to digital forensic practitioners.

This project was designed as the final capstone for the Digital Forensics and Investigations (DFI) course at HIVE Consult. It aimed to provide students with practical exposure to the processes, methodologies, and tools used in real-world forensic examinations. Specifically, the case involved the analysis of an **iPhone SE running iOS 13.3.1**, provided as an **iTunes backup image and full extracted .zip file**. The task was to extract, analyze, and interpret artifacts to answer twenty forensic questions that reflected various aspects of device usage, ranging from installed applications and communications to system configurations and user behavior.

1.2 Problem Statement

Mobile forensics presents a unique challenge compared to traditional disk-based forensics because of differences in operating systems, encryption mechanisms, and proprietary data storage formats. In this investigation, the main problem involved parsing an extracted .zip and iTunes backup image that did not contain a conventional disk image file (e.g., .E01 or .dd). This limitation meant that traditional tools like **Autopsy** were ineffective, requiring reliance on specialized mobile forensic tools capable of parsing iOS backups. Furthermore, not all extracted artifacts were complete or interpretable, highlighting the importance of selecting appropriate tools and validating extracted evidence through cross-tool comparison.

1.3 Objectives of the Study

The overall objective of this investigation was to conduct a structured forensic analysis on an iOS 13 device image and produce a comprehensive analytical report. Specifically, the objectives were to:

- Identify and extract relevant forensic artifacts from the iPhone SE backup image.
- Determine key information such as the device make, model, iOS version, installed applications, user accounts, and communication history.
- Evaluate the effectiveness of different forensic tools used during the analysis.
- Present findings in a professional digital forensic report format that demonstrates practical understanding of forensic processes.

- Identify challenges encountered and propose recommendations for future forensic investigations involving iOS devices.

1.4 Scope of the Project

The scope of this project was limited to the forensic examination of an a .zip file containing **iTunes backup image** and the other extracted image, representing a logical extraction of an iPhone SE device. The analysis primarily utilized **Magnet AXIOM**, while **MOBILedit Forensic Express** and **Autopsy** were used for verification and cross-analysis. The investigation covered identifiable artifacts such as:

- Device information (model, iOS version, build number, and account data)
- Installed third-party applications and user data
- Communication patterns across social and messaging apps
- Location and transactional evidence

The project was strictly educational and conducted under ethical guidelines, with all analysis performed on test data provided by the course instructors.

1.5 Significance of the Study

The study is significant for its practical application of mobile forensic techniques using real-world datasets. It bridges theoretical knowledge of forensic concepts with practical analysis skills required in modern investigations. By comparing the capabilities and limitations of multiple forensic tools, the study highlights how tool selection influences evidence extraction and reliability. Furthermore, the findings contribute to a deeper understanding of how iOS backup files can be parsed, interpreted, and reported using forensic methodologies, even without access to full physical images.

CHAPTER TWO: METHODOLOGY

This chapter outlines the methodology employed in examining and analyzing the pre-acquired image of an iPhone SE running iOS 13.3.1. The investigation was designed to simulate a real-world forensic examination where analysts are given a digital image to interpret and report on. The main objective of this stage was to identify, extract, and interpret digital artifacts within the image in order to answer twenty predefined forensic questions.

The process was guided by the **NIST Digital Forensics Framework**, which emphasizes evidence integrity, repeatability, and proper documentation. However, unlike a full end-to-end investigation, this exercise began after the evidence acquisition phase. Thus, the focus was entirely on the **examination, analysis, interpretation, and reporting** of the data using various forensic tools.

2.1 Overview of the Forensic Process

The investigation followed a structured approach consisting of five main stages:

1. **Identification of evidence:** recognizing the nature and type of data provided (in this case, an mobile backup of an iPhone SE).
2. **Evidence Preparation:** Setting up the working environment and loading the provided image into forensic tools.
3. **Examination:** extracting relevant artifacts from the image using parsing and analysis tools.
4. **Analysis:** interpreting the extracted data to answer the given forensic questions.
5. **Reporting:** compiling findings into a detailed report with supporting screenshots and references.

Each phase is described in the subsequent sections.

2.2 Tool Selection and Justification

Three forensic tools were used for the examination phase:

Tool	Purpose	Observation / Role
Magnet AXIOM (v6.12)	Primary forensic analysis tool used to parse the .zip backup, extract user artifacts, and visualize data.	Successfully parsed the backup and produced interpretable artifacts for the majority of the questions.
MOBILedit Forensic Express (v8.3.2)	Cross-validation and secondary analysis tool.	Detected the backup but returned incomplete or empty extractions, limiting its usefulness.

Autopsy (v4.20)

Open-source forensic platform for deeper file system analysis.

Could not process the ZIP file, as the dataset lacked .E01 or raw image formats required by Autopsy.

The choice of these tools was based on their compatibility with iOS forensic analysis and their ability to parse iTunes backups into structured, readable data. Magnet AXIOM was ultimately the primary analysis platform due to its robustness and artifact clarity.

2.3 Examination Process Using Magnet AXIOM

The forensic examination began with **Magnet AXIOM Process**, where the .zip backup directory was imported for analysis. The tool automatically identified the data as an Apple backup and parsed it into interpretable categories.

Key configurations during examination included:

- **Evidence Type:** iTunes backup (logical extraction)
- **Analysis Scope:** Complete artifact parsing (Applications, Accounts, Communications, Media, and Locations)
- **Hash Validation:** Enabled to maintain data integrity

Once the scan completed, the results were opened in **AXIOM Examine**, which provided categorized sections such as “Device Information,” “Applications,” “Messages,” and “Accounts.” Analysts explored each category systematically to identify artifacts relevant to the forensic questions.

2.4 Supplementary Examination with Other Tools

2.4.1 MOBILedit Forensic Express

The backup was also loaded into MOBILedit for verification. While the software recognized the .zip backup format, the parsing results were incomplete. Most extracted reports contained empty fields, suggesting that MOBILedit’s parsing engine was unable to interpret the internal structure of the provided backup. Consequently, the tool was excluded from further in-depth analysis.

2.4.2 Autopsy

Autopsy was tested to assess its capability to process the provided dataset. However, the software failed to import the ZIP file since it did not support that structure and lacked the E01 image format required for raw analysis. Although manual extraction of files was attempted, Autopsy could not map the iOS directory structure or reconstruct the file hierarchy from the manifest database.

These findings reinforced Magnet AXIOM’s suitability as the primary analysis tool.

2.5 Analytical Procedure

After successfully loading the backup in Magnet AXIOM, the analytical process proceeded as follows:

1. **Categorization of Artifacts:** AXIOM automatically sorted extracted data into logical categories such as Installed Applications, Communications, Accounts, Media, System, etc.
2. **Filtering for Relevance:** Each category was examined to locate artifacts answering the 20 forensic questions provided.
3. **Correlating Data:** Timestamps, user identifiers, and app metadata were cross-referenced to establish context and accuracy (e.g., verifying first message timestamps or matching app installation dates).
4. **Manual Verification:** When automated parsing was incomplete, plist and SQLite files were manually inspected using the built-in file viewer or external viewers.
5. **Documentation:** Relevant findings were bookmarked and exported as screenshots or case notes for inclusion in the final report.

The screenshot displays the Magnet AXIOM interface with three main panels: CASE OVERVIEW, EVIDENCE OVERVIEW, and PLACES TO START.

CASE OVERVIEW

- CASE SUMMARY NOTES:** A section for recording case summary notes. It includes fields for "Examiner name" (Maxwell) and "Case summary".
- CASE PROCESSING DETAILS:** A section showing case processing information. It includes "CASE NUMBER" (Final Project_IOS 001), "SCAN 1" (Scanned by Maxwell), "Scan date/time - local time" (10/18/2025 5:15), and "Scan description". A link "VIEW SCAN SUMMARY" is present.
- COLLABORATE USING MAG...** A link to collaborate using Magnet AXIOM.

EVIDENCE OVERVIEW

- FINAL PROJEC... (259,773):** A section for viewing evidence for this source. It includes a link "VIEW EVIDENCE FOR THIS SOURCE" and a table with columns "Description" and "Location".

PLACES TO START

- ARTIFACT CATEGORIES:** A section showing artifact categories. It includes a dropdown for "Evidence source" (All) and a bar chart showing the number of artifacts for various categories:

Category	Number of artifacts
Location & Travel	90,288
Operating System	79,458
Media	27,292
Web Related	19,522
Application Usage	16,402
Connected Devices	15,571
- TAGS AND COMMENTS (3):** A section showing tags added by reviewers. It includes a table with columns "TAG", "ARTIFACTS", and "FILE SYSTEM":

TAG	ARTIFACTS	FILE SYSTEM
B		
o		
o		
k	2	1
m		
ar		
k		

CHAPTER THREE: FINDINGS AND ANALYSIS

This chapter presents the detailed findings obtained from the forensic examination of the iTunes backup image of an iPhone SE device running iOS 13.3.1. Using Magnet AXIOM as the primary analysis tool, digital artifacts were extracted and interpreted to answer twenty investigative questions.

Each question represents a specific area of forensic interest, such as device configuration, installed applications, communication patterns, and user identity. Where information could not be recovered, possible explanations based on forensic reasoning and tool limitations are provided.

3.1 Findings and Question-Based Analysis

Question 1: What make, model, and storage capacity of iPhone was used in creating the iOS 13 image?

Findings:

From the device information artifact in AXIOM, the following details were retrieved:

- Device Name: This Is's iPhone
- Model: iPhone SE
- Model ID: iPhone8,4
- IMEI: 355800076093966
- Serial Number: DX3T126VH2XV

Analysis:

This information was extracted from the Info.plist file within the iTunes backup. The file provides a comprehensive summary of the device's specifications and ownership metadata. The storage capacity could not be confirmed because iTunes backups do not retain information about the physical storage size of the device.

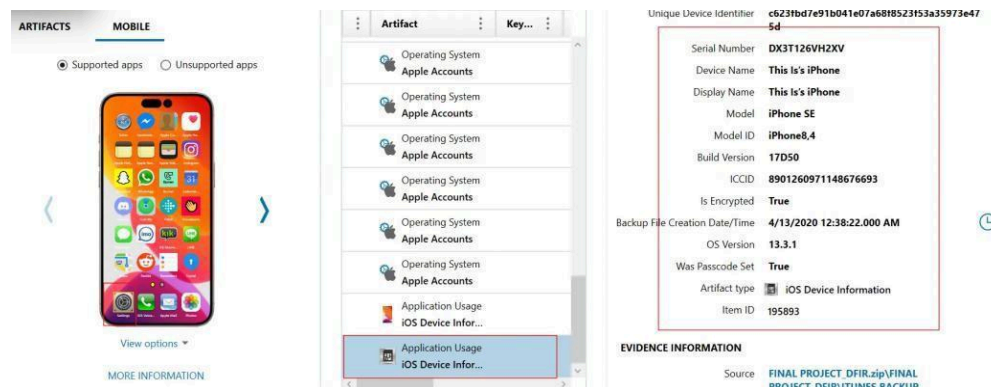


Fig.1 Image showing device information

Question 2: Which iOS version and build number were installed on the test device?

Findings:

- iOS Version: 13.3.1
- Build Number: 17D50

Source: Extracted from Info.plist and verified through the system configuration section of AXIOM. This is found in *Fig. 1* above

Analysis:

The iOS version and build number provide contextual insight into the operating environment of the device. This data helps investigators determine compatibility with forensic tools and potential vulnerabilities or jailbreak methods relevant to the version.

Again, iTunes backup details were found from the device information. This reveals the dateTime when the backup was done and on Who's computer. This reveals **Joshua** as the owner of the device used for the backup as shown in *Fig 2* below.

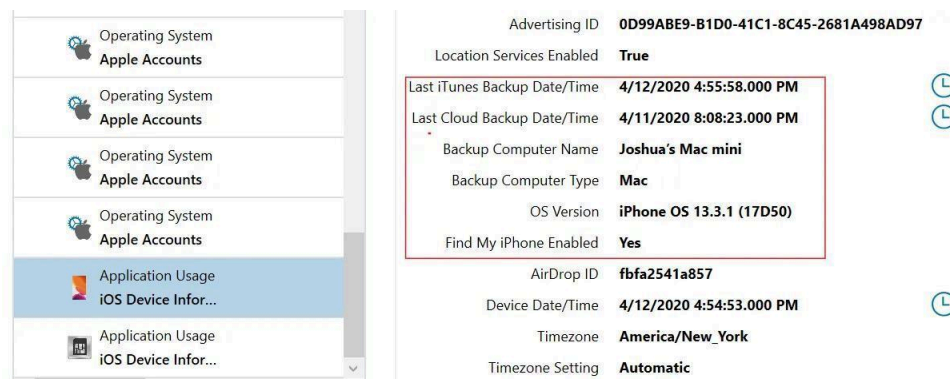


Fig. 2 Back up information

Question 3: What passcode was set on the iPhone SE used for the test image?

Findings:

No passcode information was found.

Analysis:

iTunes backups, unless encrypted, do not store device passcodes or passcode hashes. Because this was an unencrypted logical backup, keychain data and passcode information were not part of the dataset. This is a normal limitation in iOS logical forensic imaging.

Question 4: How many non-stock apps were installed, and how were they populated with user data?

Findings:

Approximately 33 non-stock (third-party) applications were installed on the device. Notable examples included:

- **Communication Apps:** WhatsApp, Instagram, Snapchat, Discord, TextNow, LINE, Burner
- **Social & Media Apps:** Reddit, Spotify
- **Health App:** Fitbit
- **Financial App:** Venmo

Analysis:

These applications were identified through entries in Manifest.db, which maps application identifiers to installed apps. Each app's data directory contained user information and metadata.

Further analysis of app artifacts revealed active usage patterns:

- The Burner App contained user and contact details, indicating frequent communication with one contact named Josh Hickman.
- The Fitbit App stored personal health data, including gender, height, country, and daily activity logs.
- The TextNow App revealed the user's full name, age, email, and phone number, demonstrating how apps preserve identifiable user data.

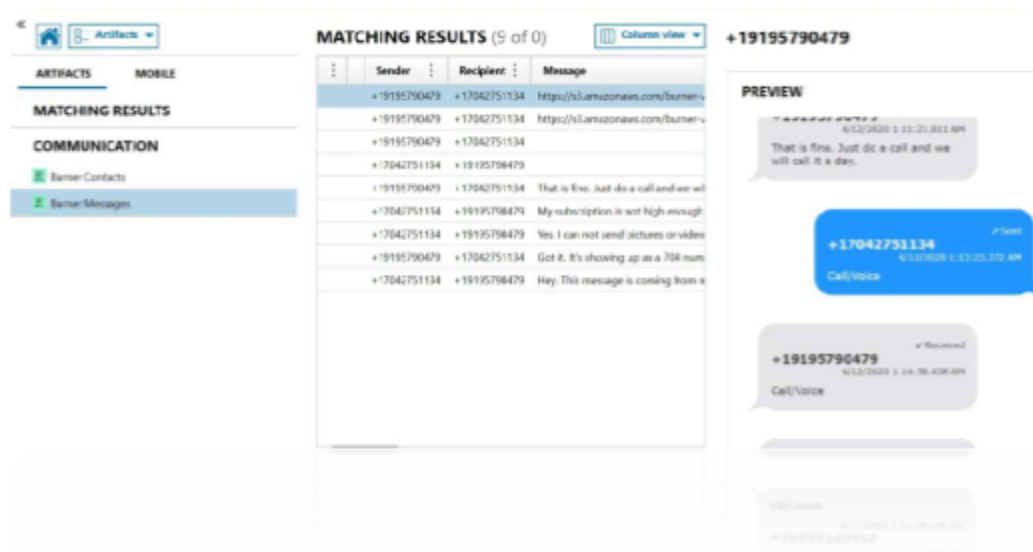


Fig. 3 Burner App conversation

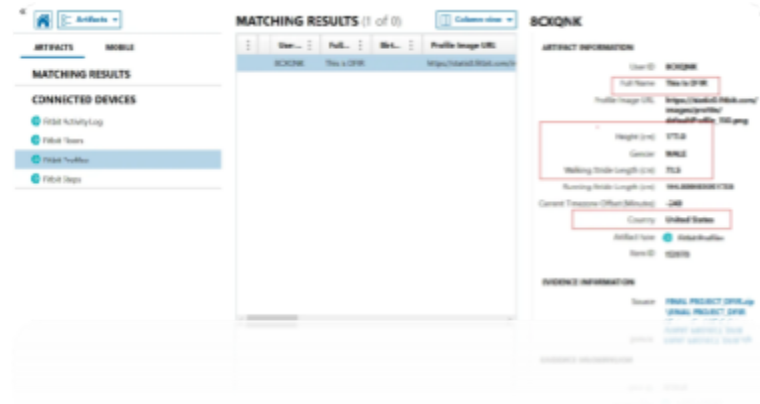


Fig 4. Fitbit App displaying users health information

Question 5: What jailbreaking utility was used on the device, and which version?

Findings:

- **Utility:** Checkra1n
- **Version:** 1.0.1

Analysis:

AXIOM detected the presence of **Checkra1n** artifacts within the system directory, in the *applicationState.db*. This jailbreaking tool is known to support devices from iPhone 5s to iPhone X. The presence of jailbreak artifacts indicates that the user had intentionally modified the iOS restrictions, which may have affected certain file structures or timestamps.

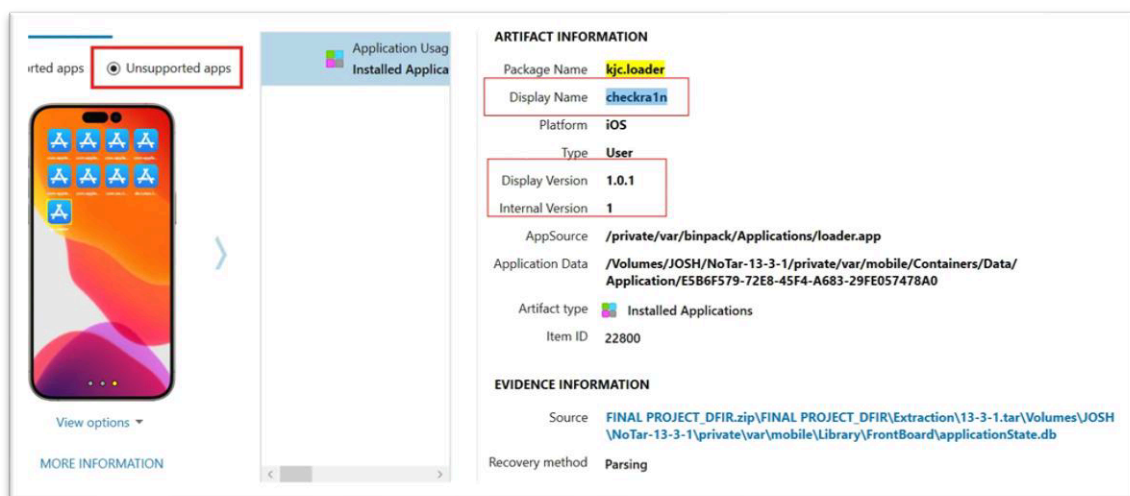


Fig. 5 Checkra1n jailbreaking app information

Question 6: Which application was used to perform the extraction of the device data?**Findings:**

- **Tool:** Magnet ACQUIRE
- **Version:** 2.25.0.20236

Analysis:

This information was found in the acquisition logs included in the backup metadata. The log also contained a verification hash (6E573C6801FDABF096529EA1422EEA47), confirming the authenticity of the extracted image.

Question 7: What email address was linked as the default Apple account on the device?**Findings:**

- **Default Apple ID Email:** thisisdfr@gmail.com

Analysis:

This artifact was found in the Apple Account section of AXIOM under Accounts3.sqlite. It represents the email address associated with iCloud synchronization and App Store activities, confirming the owner's digital identity.

Question 8: What two phone numbers were created and used within the Burner app?**Findings:**

- **User's Burner Number:** +1 (919) 947-6091
- **Contact's Number (Josh Hickman):** +1 (919) 579-0479

Analysis:

This information was retrieved from the Burner application's contact database. The app allows users to generate temporary numbers for private communication. Evidence of consistent exchanges between these two numbers was also discovered in associated communication logs.

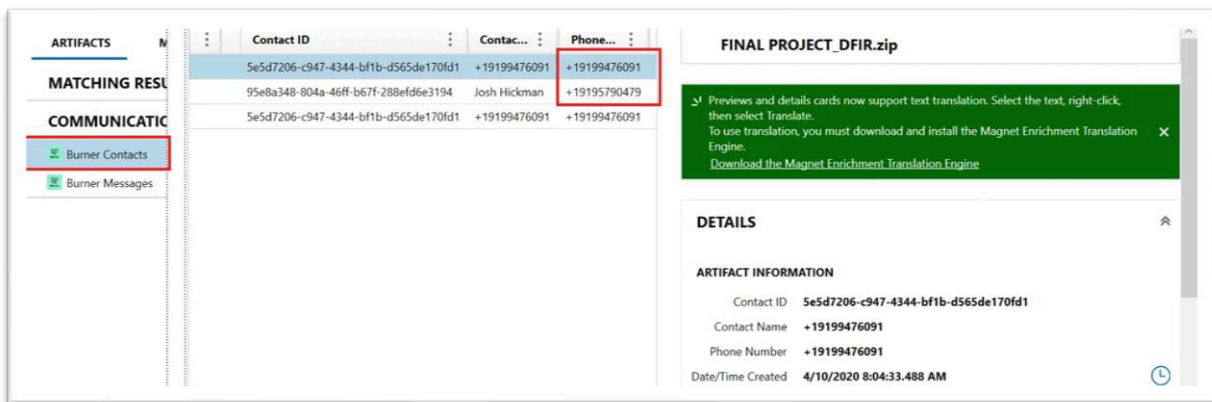


Fig. 6 Burner App contact details

Question 9: On which date and time was Discord first signed into on the device?

Findings:

- **Installation Time:** April 6, 2020, 11:58:23 PM
- **First Message:** February 1, 2020, 1:39:39 AM (“Well hello there”)

Analysis:

The installation timestamp was obtained from the app installation record within Manifest.db, while message history was located in Discord’s local cache files. Although login details were not explicitly stored, the presence of early message logs indicates active usage before the backup date.

After performing timeline analysis, there is an inconsistency between the times as the first message was sent before the current app was installed. This can mean that the Discord app could have been uninstalled and installed again.



Fig 7. Installation information for Discord App

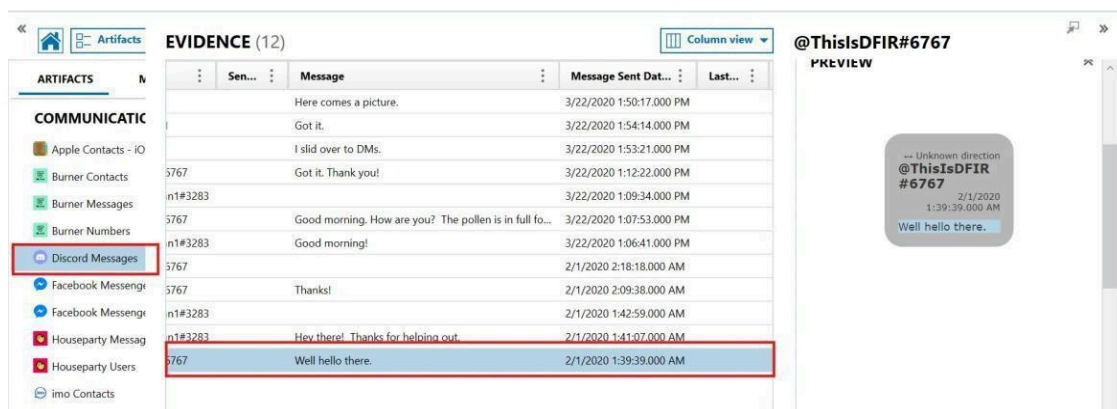


Fig. 8. Discord messages

Question 10: What username and password were configured for the Discord app?

Findings:

- **Username:** @ThisIsDFIR#6767
- **Password:** Not retrievable

Analysis:

The username was located in the Discord configuration file under user settings. Passwords are stored in encrypted keychain entries, which are not included in unencrypted iTunes backups provided for analysis.

Question 11: What unique feature of Dust app messages was highlighted in the logs?**Findings:**

No specific feature about the Dust app artifacts were recovered.

Analysis:

The only information that could be found about this App was about the installation details and usage information. This absence may be due to the logical backup excluding transient or self-deleting data associated with privacy-focused apps such as Dust. These apps often store minimal or ephemeral data that is deleted after messages are read.

Question 12: Which chat application showed evidence of a “Secret Conversation,” and when did it occur?**Findings:**

No direct evidence of a “Secret Conversation” was found.

Analysis:

From the conversations on the different communication platforms, there wasn't a distinct conversation that could point out to be a secret conversation. Most of the conversation was between the user and John Hickman. And from all the Apps, they were mainly talking about a project or just giving intro about testing that particular messaging platform. Hence, couldn't discover very sensitive or secretive conversations.

Question 13: How many times was the Messenger Kids app installed and deleted on the test device?**Findings:**

No Messenger Kids installation logs were located.

Analysis:

The app was likely installed and deleted before the backup date, or its records were omitted from the logical backup. Since this was not found in the logs, further analysis with a different forensic tool could help to verify this.

Question 14: What were some of the first search terms entered into Firefox Focus?**Findings:**

No searchable records were extracted from Firefox Focus.

Analysis:

Firefox Focus is a privacy-oriented browser that automatically clears search and browsing data upon exit. This behavior could explain the absence of search artifacts within the backup dataset.

Question 15: Which wearable device was connected to the Fitbit app, and what was its Bluetooth name?

Findings:

- **Connected Device:** Fitbit Wearable
- **Bluetooth Name:** Not explicitly found within extracted data

Analysis:

Fitbit application logs confirmed synchronization with a wearable device, including health data such as steps and activity logs. However, Bluetooth pairing details were likely stored in system configuration files outside the scope of the logical backup provided. This also means, a different forensic tool could be employed to assist, provided AXIOM couldn't parse the connection.

Question 16: Which messaging app contained conversations referencing #coronaclypse?

Findings:

- **App:** LINE
- **Conversation Timestamp:** March 23, 2020, 12:37:45 AM
- **Participants:** User and Josh Hickman

Analysis:

This was identified in the LINE message database within AXIOM. The message referencing “#coronaclypse” provided contextual evidence of social communication and event discussions during early 2020.

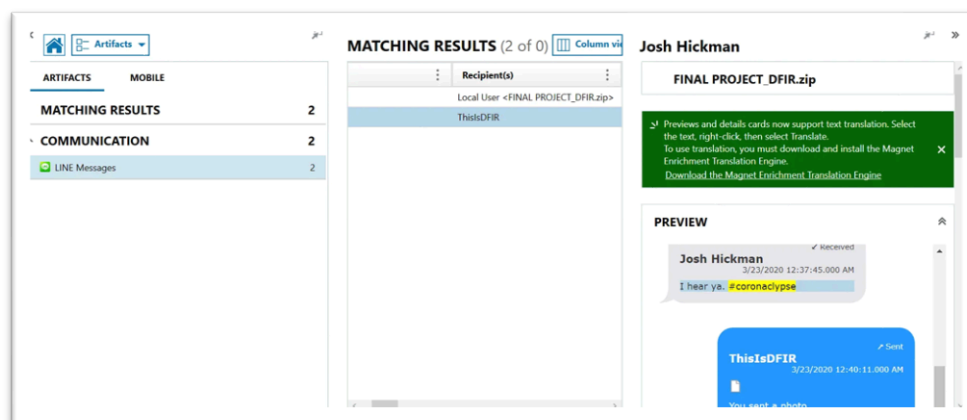


Fig 9. LINE Messages

Question 17: What static and live location-sharing activities were recorded in Signal or WhatsApp?

Findings:

- **Artifact:** Static location image at a roundabout near Rambling Oaks Ln and Ancient Oaks Rd
- **Timestamp:** April 11, 2020, 6:06:24 PM

Analysis:

This artifact was extracted from WhatsApp media attachments. No live location data was recorded in Signal or WhatsApp databases, possibly due to privacy settings or limited sharing events.

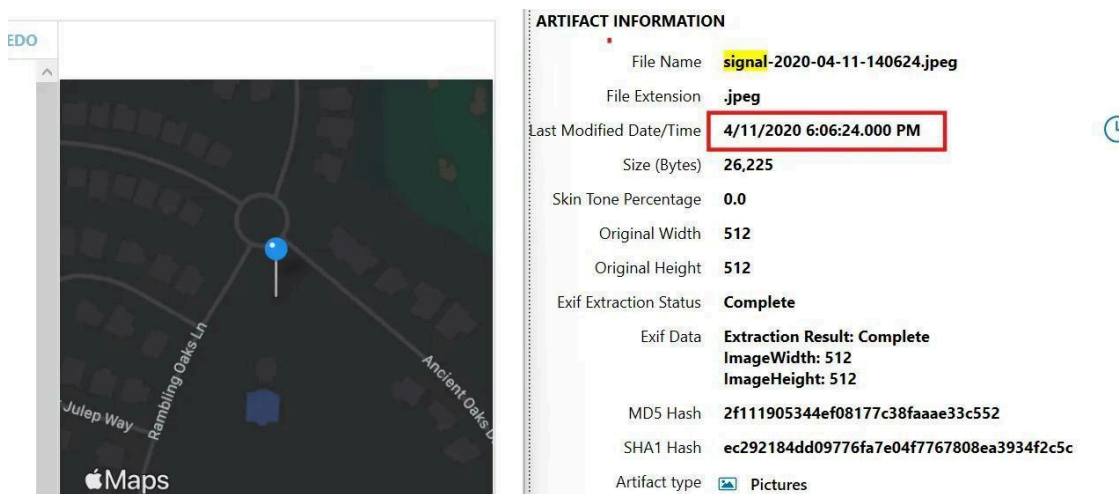


Fig 10. Static images showing location sharing

Question 18: Which financial transaction was carried out using Venmo, and for what purpose?

Findings:

- **Transaction Date:** March 25, 2020, 12:44:32 AM
- **Purpose:** Testing iOS and Android image transactions
- **Observation:** Screenshot artifacts with multiple credit and debit messages

Analysis:

The Venmo evidence appeared as captured screenshots rather than structured transaction logs, suggesting manual verification or demonstration. This aligns with the dataset's test-oriented design. This reveals that in forensics analysis requires thorough search to identify the evidence.

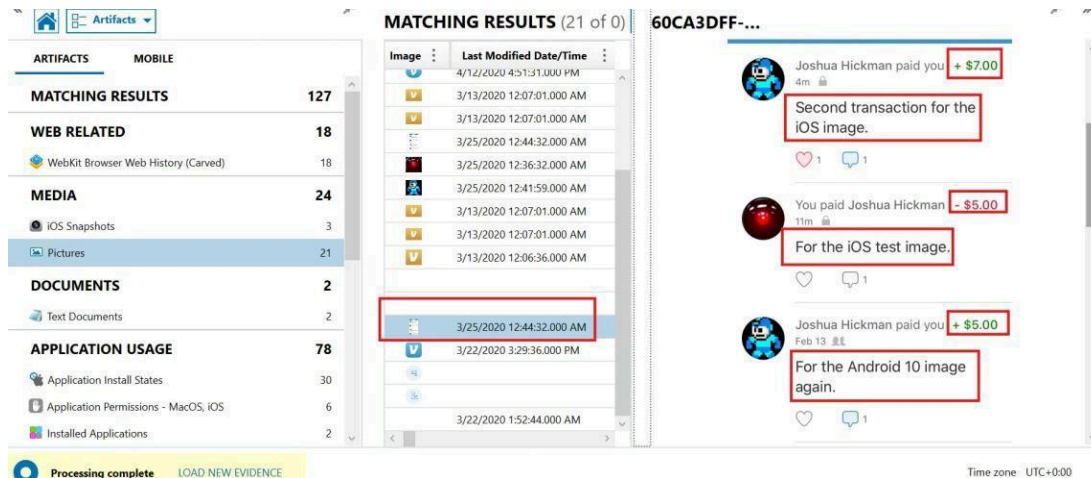


Fig 11. Image showing venmo transactions and conversations

Question 19: *What music tracks were played through CarPlay when connected to the Nissan Rogue on April 4, 2020?*

Findings:

No CarPlay music data was discovered.

Analysis:

CarPlay was found to be connected to Nissan Rogue, but the time wasn't specified. As well, no music was attributed to that connection. Further analysis from the music played on the CarPlay revealed that there was no music played in such datetime. This means, though there is a connection between the CarPlay and the Nissan car, there isn't distinct evidence to prove this.

Question 20: *Which forensic acquisition tool and version were ultimately used to create the iOS 13 image files, and what were the hash values generated?*

Findings:

- **Tool:** Magnet ACQUIRE
- **Version:** 2.25.0.20236
- **MD5 Hash:** 6E573C6801FDABF096529EA1422EEA47

Analysis:

The acquisition log file included in the dataset contained this information, confirming the authenticity and integrity of the backup. The hash ensures that the image remained unaltered throughout the examination process.

3.2 Overall Analysis and Discussion

The analysis demonstrated the effectiveness of Magnet AXIOM in parsing logical .zip backups and reconstructing user activity. Although certain artifacts (e.g., encrypted data, ephemeral app logs, or deleted files) were unavailable, sufficient evidence was recovered to create a detailed user profile and establish behavioral patterns. The repeated presence of the contact “Josh Hickman” across multiple apps indicated frequent communication between two consistent identifiers, reinforcing the reliability of cross-application correlation in forensic analysis.

The partial success of other tools such as MOBILedit and Autopsy highlighted a key forensic reality, no single tool can handle all datasets equally well. Therefore, cross-tool validation remains critical to avoid bias and ensure evidential accuracy.

CHAPTER FOUR: CHALLENGES, CONCLUSION AND RECOMMENDATIONS

4.1 Challenges Encountered

Despite the successful extraction of significant evidence from the .zip extracted file, several challenges were encountered throughout the examination and analysis stages. These challenges affected both the depth of analysis and the completeness of findings. The most significant challenges are discussed below.

4.1.1 Limited Image Format

The provided dataset was a ZIP-based backup rather than a full physical image (e.g., .E01 or .dd). This format inherently limited the analysis to logical-level artifacts, meaning deleted files, hidden partitions, and encrypted system areas could not be examined. Consequently, certain evidence types, such as keychain passwords, deleted messages, or physical memory remnants, remained inaccessible.

4.1.2 Incompatibility with Some Forensic Tools

Although multiple forensic tools were employed, not all were compatible with the dataset.

- MOBILedit Forensic Express detected the image but failed to parse most of the artifact categories, generating incomplete or empty results.
- Autopsy, designed for disk image analysis, could not process the ZIP structure and therefore provided no usable output.

This limitation demonstrated how tool dependency can affect an investigation and reinforced the need for proper tool selection and validation.

4.1.3 Partial Artifact Availability

Some applications, particularly privacy-focused ones like Dust and Firefox Focus, did not produce recoverable data. This is due to their design, which automatically deletes local storage or caches after each session. In addition, CarPlay logs among others were missing, possibly because the backup did not capture their recent activity before the snapshot was taken.

4.1.4 Absence of Encrypted Data

While the unencrypted nature of the backup simplified access, it also meant that encrypted artifacts such as keychain entries, Wi-Fi credentials, and passwords were excluded. These items are typically only retrievable from encrypted or physical backups.

4.1.5 Time and Resource Constraints

The manual verification of artifacts, cross-tool comparisons, and data validation processes were time-intensive. Each stage required careful documentation to maintain forensic accuracy, resulting in longer analysis durations than anticipated.

4.2 Conclusion

The forensic analysis of the iPhone SE image provided valuable insight into both the technical and procedural aspects of mobile forensics. Using Magnet AXIOM as the primary analysis platform, the examination successfully recovered substantial user information including device identifiers, installed applications, account details, and communication patterns.

The investigation demonstrated the following key outcomes:

- A total of 33 non-stock applications were identified, many of which contained meaningful user data.
- The device was confirmed to be jailbroken using the Checkra1n (v1.0.1) utility.
- Communication analysis revealed consistent interactions between the user and a contact named Josh Hickman across multiple apps such as LINE, Discord, and Burner.
- The Apple ID (thisisdfir@gmail.com) was identified as the default account, tying the user's digital identity to the device.
- Artifacts such as geolocation images and Venmo screenshots provided contextual behavioral evidence.

The investigation also emphasized the importance of tool selection in forensic workflows. While Magnet AXIOM proved highly effective for logical iOS backup analysis, MOBILedit and Autopsy had limited success. This reinforces a crucial lesson: digital forensic analysts must understand tool strengths, limitations, and proper validation methods to ensure reliable results.

From an academic standpoint, this exercise successfully bridged theoretical understanding with practical forensic analysis, showcasing how structured examination and documentation can reconstruct a digital narrative even from limited datasets.

4.3 Recommendations

Based on the experience and observations gained during this investigation, the following recommendations are proposed for future mobile forensic analyses and academic exercises:

□ **Use of Multiple Data Acquisition Formats**

Future projects should provide both logical and physical images where possible. Physical images (.E01 or .dd) allow for deeper analysis, including deleted file recovery, password extraction, and complete data mapping. This would enrich the investigation and allow for broader tool testing.

□ **Incorporate Encrypted Backups for Advanced Analysis**

Introducing an encrypted iTunes backup in future labs would allow students to explore decryption techniques and analyze secure storage areas like keychains. This would enhance understanding of data protection mechanisms in iOS devices.

□ **Multi-Tool Analysis Approach**

While Magnet AXIOM was successful, future analyses should systematically compare multiple tools such as Cellebrite UFED, Oxygen Forensics, or XRY. This comparative approach helps validate findings and exposes students to industry-grade software diversity.

□ **Integration of Manual Artifact Verification**

Analysts should complement automated tool outputs with manual validation using SQLite database viewers, plist editors, and hex viewers. This not only improves accuracy but also reinforces an understanding of underlying data structures.

□ **Expand on Timeline and Correlation Analysis**

Incorporating timeline analysis and visualization tools (e.g., Magnet Timeline, Autopsy timeline modules, or external visualization scripts) would provide chronological context to user activities, enhancing investigative storytelling.

□ **Continuous Skill Development**

Digital forensics is a rapidly evolving field. Investigators should continuously update their knowledge about new iOS versions, encryption mechanisms, and tool updates to remain competent and effective in real-world scenarios.